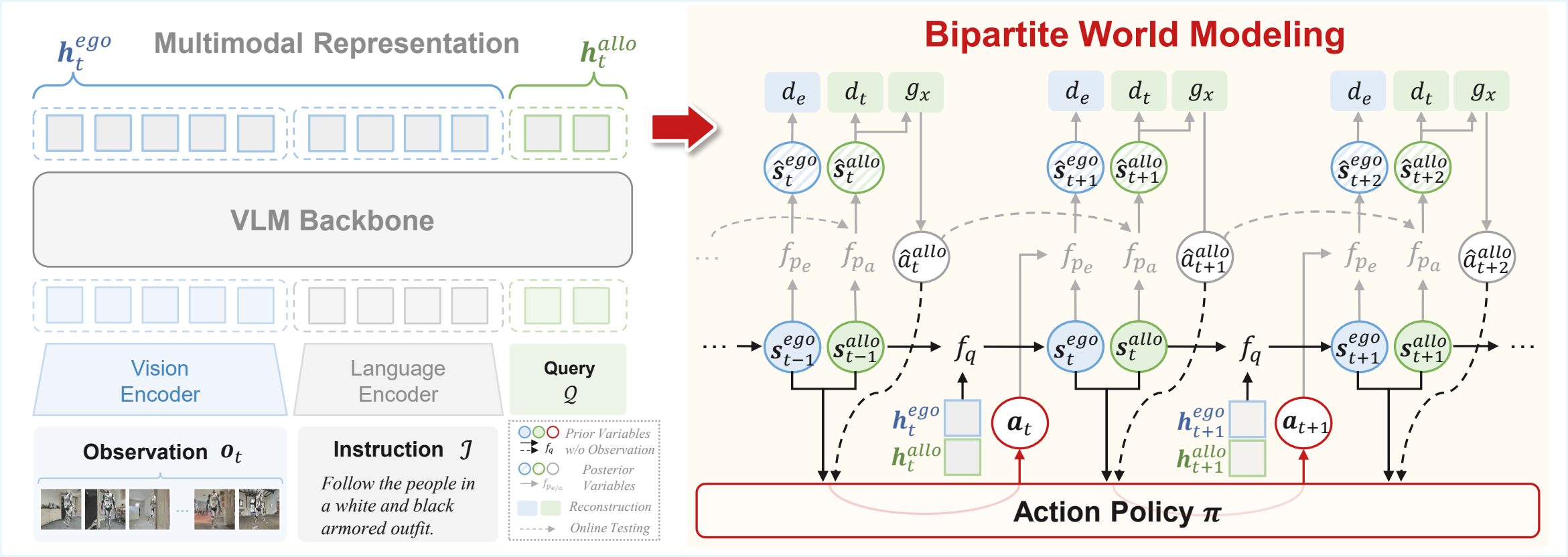

h_t^{ego}
Multimodal Representation
h_t^{allo}
VLM Backbone
Vision Encoder
Language Encoder
Query $\mathcal{Q}$
Observation o_t
Instruction $\mathcal{I}$
Follow the people in a white and black armored outfit.
Prior Variables w/o Observation
Posterior Variables
Reconstruction
Online Testing
Bipartite World Modeling
d_e d_t g_x
$\hat{s}_t^{ego}$ $\hat{s}_t^{allo}$
f_{p_e} f_{p_a}
$\hat{a}_t^{allo}$
s_{t-1}^{ego} s_{t-1}^{allo}
f_q
s_t^{ego} s_t^{allo}
h_t^{ego} h_t^{allo}
a_t
$\hat{s}_{t+1}^{ego}$ $\hat{s}_{t+1}^{allo}$
$\hat{a}_{t+1}^{allo}$
s_{t+1}^{ego} s_{t+1}^{allo}
h_{t+1}^{ego} h_{t+1}^{allo}
a_{t+1}
$\hat{s}_{t+2}^{ego}$ $\hat{s}_{t+2}^{allo}$
$\hat{a}_{t+2}^{allo}$
Action Policy π